

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

September 2019

EC 308//308.01/347 Antenna & Wave Propagation

Date: 16.09.2019, Monday Time: 10.00 a.m. To 1.00 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION -I

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|-----|----|---|---|
| Q-1 | a. | If the impedance of dipole antenna is 73 ohm then what is the impedance of its complementary antenna? | 2 |
| | b. | If dipole is horizontally polarized than its complementary antenna is _____polarized. | 1 |
| | c. | Two identical transmitting and receiving antennas are located at a distance of 2km. Power transmitted is 30dB at 1GHz and received power is 10dB, the approximate gain of each antenna is | 2 |
| | d. | Discuss advantages and disadvantages of microstrip patch antenna. | 2 |
| Q-2 | a. | Determine the distance from short dipole operating at 1MHz at which radiation field is equal to the induction field. | 2 |
| Q-2 | b. | Derive the E field equation of array two point sources with equal amplitude and opposite phase also draw radiation pattern. | 6 |
| | c. | Briefly discuss n Elements Uniform Linear Arrays of point sources – broad side case | 6 |

OR

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| Q-2 | b. | Write the statement of pattern multiplication theorem. Explain it in detail by using any example. | 6 |
| | c. | Explain in detail different types of propagations. | 6 |
| Q-3 | a. | Explain Rumsay's principle for frequency independent antenna. | 2 |
| | b. | If circumference of helical antenna is 0.5 meters, find out max and min frequency and fractional BW. | 2 |
| Q-3 | c.(1) | A pyramidal horn with E plane aperture $a_E=10\lambda$ is fed by rectangular waveguide with TE ₁₀ mode. Let $\delta=0.2\lambda$ in E plane and 0.375λ in H plane. Find following parameters. Length of horn, H plane aperture flare angle in E plane and H plane. HPBW for E plane and H plane. | 4 |
| | c.(2) | Design log periodic antenna which operates on 50MHz to 250MHz and provide 8 dB gain. ($S=0.17\lambda, k=1.1$) | 6 |

OR

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| Q-3 | c.(1) | Explain SWR measurement method for antenna impedance measurement. | 5 |
| | c.(2) | Explain yagi uda array in detail. | 5 |

SECTION II

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|-----|----|---|---|
| Q-4 | a. | The measured HPBW's of an antenna in the two orthogonal planes are 10° and 20° . Antenna efficiency 100%. The approximate gain of the antenna is :_____ . | 1 |
|-----|----|---|---|

	b.	If antenna directivity and antenna gain are equal, then antenna efficiency is _____%.	1
	c.	The radiation resistance of antenna is 80 ohm and loss resistance is 10 ohm. What is the directivity, if power gain is 10?	1
	d.	Calculate the front to back ratio of an antenna in dB which radiates 3 kW in its most optimum direction and 500 watts in opposite direction.	1
	e.	The radiation resistance of half wave dipole is _____.	1
	f.	Explain Skip Zone, Radiation pattern	2
Q-5	a.	Explain different Feed Mechanism used in parabolic Reflector antenna systems.	5
	b.	Design rectangular patch antenna at 10GHz, dielectric constant 10.2 and thickness of dielectric material is 0.127cm.	5
	c.	Derive equation of received power of antenna (FRISS formula).	4
		OR	
Q-5	a.	Draw and explain different types of reflectors	5
	b.	Explain Babinet's principle and discuss it for complementary antenna.	5
	c.	Explain different feed techniques used in patch antenna	4
Q-6	a.	Design Rhombic antenna (leg length, tilt angle, height above ground) at frequency 10 GHz and β is 10° .	2
Q-6	b.(1)	Sketch the helical geometry with its associated dimension showing relationship between circumferences, spacing, turn length and pitch angle of helix. Also discuss practical design consideration.	6
	b. (2)	Draw different types of horn antenna and explain them in detail.	6
		OR	
Q-6	b.	Starting from retarded currents derive an expression for E and H components for a short dipole antenna if the spherical system is defined in r , θ and Φ .	12